

Sales Analysis Dashboard Using Power BI

Prepared By: Banty Raj



Introduction

Project Overview

The objective of this project is to analyze sales performance using Power BI and derive meaningful business insights through descriptive, comparative, and predictive analytics.

Objectives

- Understand sales performance.
- Analyze customer and product behavior.
- Identify trends and patterns.
- Detect outliers and anomalies.
- Generate predictive insights using Power BI visuals.

Dataset Overview

Dataset Information

- Total Records: 9,994
- Number of Columns: 21
- Data Source: Sales Dataset

Key Columns

- Order Date
- Customer Name
- Segment
- Region
- Category
- Sub-Category
- Sales
- Quantity
- Discount
- Profit

Insert Dataset Screenshot

Requirement Analysis

Business Requirements

- Perform basic exploration of the data.
- Conduct comparative analysis.
- Generate predictive insights.
- Build interactive dashboards.
- Present actionable recommendations.

Methodology

Process Followed

1. Data Collection
2. Data Cleaning
3. Data Transformation
4. Data Modeling
5. DAX Measure Creation
6. Dashboard Development
7. Insight Generation

Data Quality Analysis

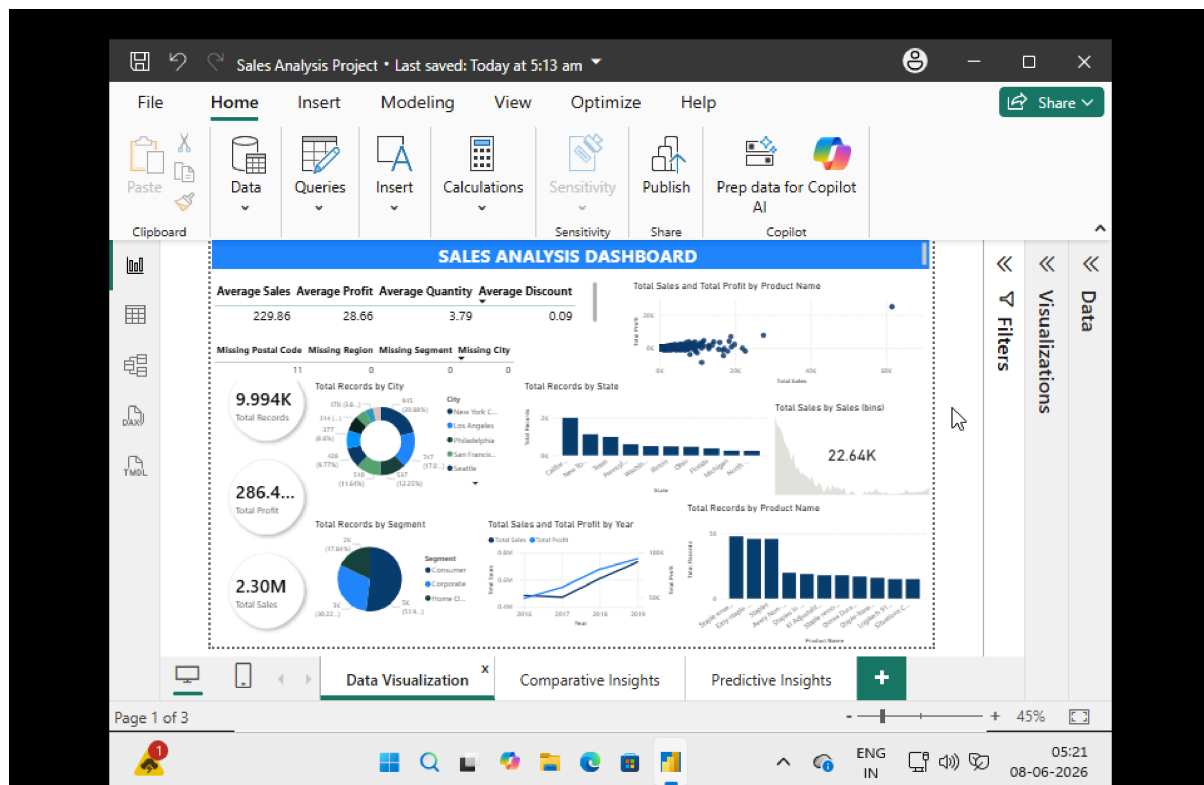
Missing Values Analysis

- Checked all columns for missing values.
- Postal Code contains limited missing values.
- Most fields are complete and reliable.

Key Insight

Dataset quality is suitable for analysis.

Basic Exploration Dashboard



Key Metrics

- Total Records
- Average Sales
- Average Profit
- Average Quantity
- Average Discount

Insights

- Dataset contains nearly 10,000 records.
- Average transaction value provides a baseline for business performance.

Categorical Distribution Analysis

Insert:

- Category Distribution Chart
- Segment Distribution Chart
- Region Distribution Chart

Insights

- Consumer segment contributes the highest transaction volume.
- Certain categories dominate overall sales activity.

Top 10 Analysis

Insert:

- Top 10 States Chart
- Top 10 Cities Chart

Insights

- Sales activity is concentrated in a few locations.
- Major cities contribute a large share of transactions.

Trend Analysis

Insert Sales Trend Line Chart

Insights

- Sales fluctuate over time.
- Seasonal spikes indicate peak purchasing periods.

Outlier Analysis

Insert:

- Sales vs Profit Scatter Plot
- Discount vs Profit Scatter Plot

Insights

- Several transactions generate unusually high profits.
- High discounts often correspond with lower profits.

Distribution Analysis

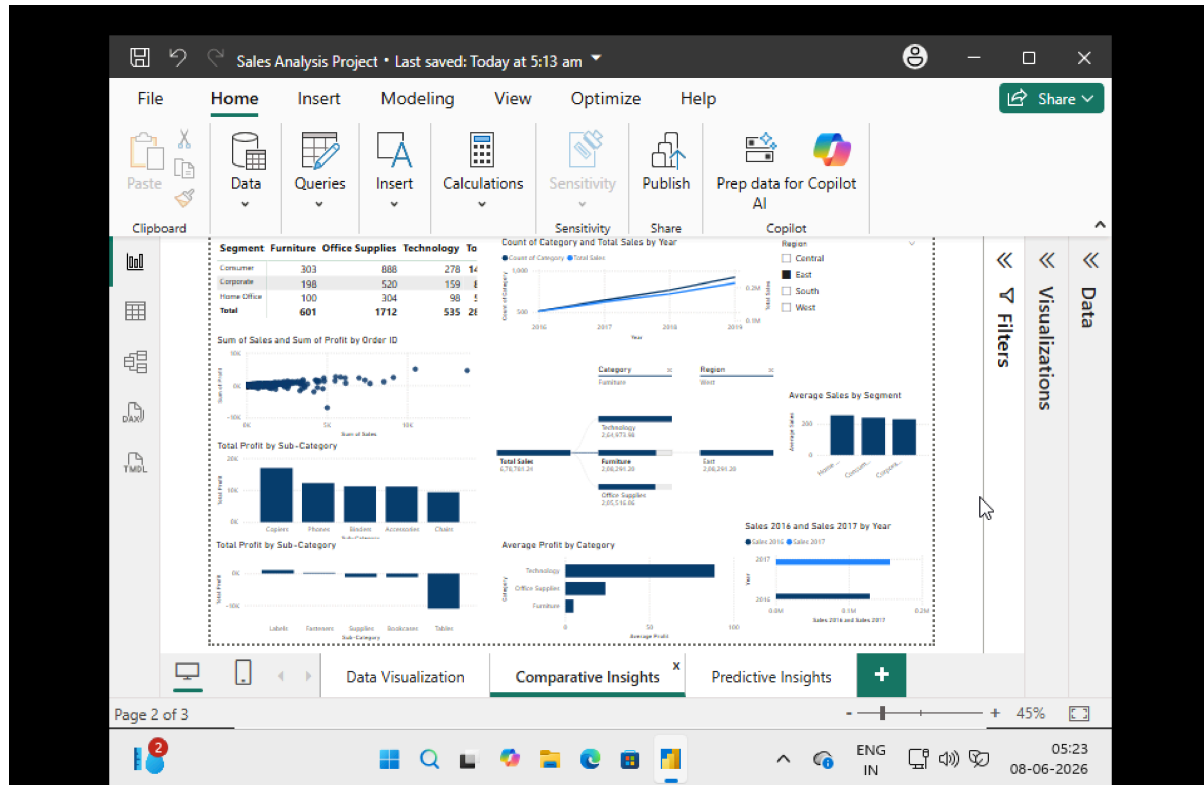
Insert:

- Sales Histogram
- Profit Histogram
- Quantity Histogram

Insights

- Sales distribution is positively skewed.
- Most orders are small while a few are extremely large.

Comparative Insights Dashboard



Comparative Analysis Objectives

- Compare categories.
- Evaluate segment performance.
- Examine category relationships.

Top and Bottom Performing Categories

Insert:

- Top 5 Categories Chart
- Bottom 5 Categories Chart

Insights

- Certain categories consistently generate high profit.
- Some categories require strategic improvement.

Category Trends Over Time

Insert Multi-Line Category Trend Chart

Insights

- Technology shows strong growth.
- Some categories experience seasonal variations.

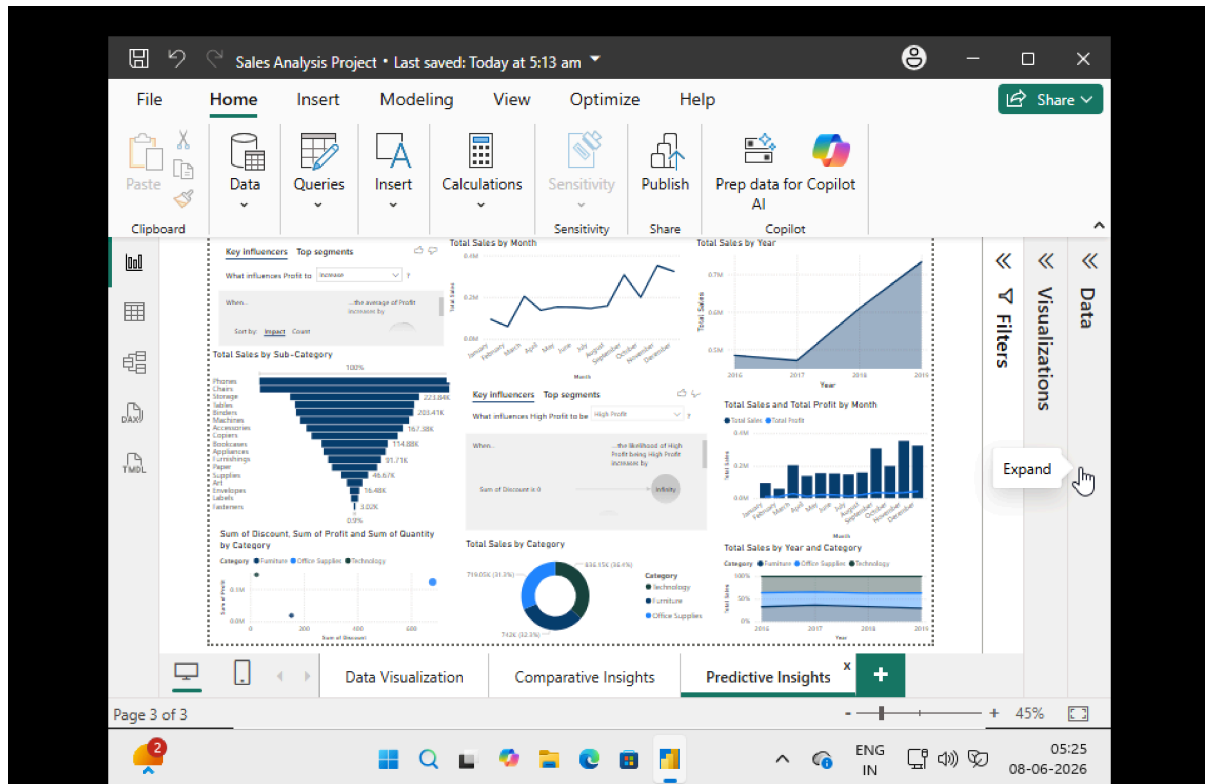
Decomposition Tree Analysis

Insert Decomposition Tree Screenshot

Insights

- Region, Category, and Segment significantly impact profitability.
- Certain combinations drive business success.

Predictive Insights Dashboard



Objective

Use predictive and advanced analytical techniques to identify future opportunities.

Key Influencers & Forecasting

Insert:

- Key Influencers Visual
- Forecasting Chart

Insights

- Product category and discount are major drivers.
- Forecast indicates continued sales growth trends.

Business Recommendations

Recommendations

1. Focus on high-performing categories.
2. Reduce excessive discounting.
3. Improve performance of low-profit products.
4. Expand operations in high-performing regions.
5. Monitor outlier transactions regularly.

Conclusion

Summary

This project successfully analyzed sales data using Power BI dashboards and advanced analytical techniques.

Major Findings

- Strong sales concentration in specific categories and regions.
- Clear seasonal sales patterns.
- Outliers significantly influence profitability.
- Predictive analysis indicates future growth opportunities.

Final Conclusion

Power BI provides a comprehensive platform for transforming raw sales data into actionable business intelligence that supports strategic decision-making.